

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	30.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Dysuria;Lower pelvic pain

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	39.0	%	20 - 40
Wbc	8850.0	cells/μL	4000 - 11000
Polymorphs	62.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.0	mg/dL	3.5 - 7.2
Blood Urea	23.0	mg/dL	7 - 20
Cue Pus Cells	10.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clarithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (e.g., 400 mg IV daily for a 70 kg adult)

Precautions: Monitor serum CPK levels for myopathy; adjust dose if creatinine clearance <30 mL/min; avoid concurrent high-dose statins; assess for hypersensitivity; patient has normal renal function (creatinine 0.7 mg/dL) so standard dosing is appropriate.

Rationale: Enterococcus faecium in this case is predicted to be resistant to clarithromycin and only sensitive to daptomycin. Daptomycin is active against VRE and provides reliable bactericidal activity for this gram-positive infection.

Antibiotic Drug History

Clinical Summary

Infection Summary

- **Type:** Recurrent lower-tract urethritis (urinary tract infection).
- **Predicted pathogen:** Enterococcus faecium, a gram-positive coccus with a high likelihood of vancomycin resistance (VRE).

Antibiotic Susceptibility

- **Resistant:** Clarithromycin (predicted).
- **Sensitive:** Daptomycin (predicted to be the only effective agent).

Recommended Therapy

- **Drug:** Daptomycin 6 mg/kg IV once daily (e.g., 400 mg IV daily for a 70-kg adult).
- **Rationale:** Daptomycin retains potent bactericidal activity against VRE, including E. faecium, and is the sole

agent predicted to be susceptible. Prior use of clindamycin and vancomycin provides no coverage for this organism.

Key Precautions & Monitoring

- Check baseline and periodic serum creatine phosphokinase (CPK) to detect myopathy; hold or adjust if CPK rises markedly.
- Dose adjustment is required only if creatinine clearance falls <30 mL/min; the patient's current creatinine (0.7 mg/dL) permits standard dosing.
- Avoid concurrent high-dose statins due to increased myopathy risk.
- Observe for hypersensitivity reactions.

Clinical Action

Initiate daptomycin as above, monitor CPK and renal function, and reassess after 48-72 h for clinical response and any adverse effects. Adjust therapy if culture results diverge from the prediction.